Groundwater Data Pipeline & Models

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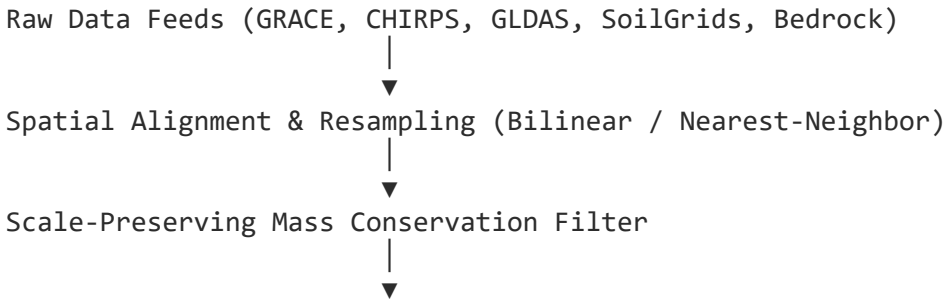
User prompt: summarise

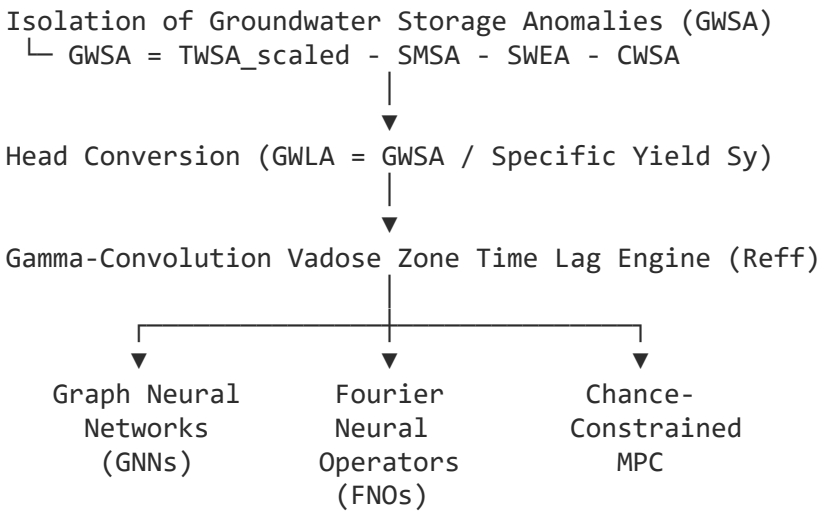
Response: This catalog outlines an AI-driven hydroinformatic pipeline for modeling macro-to-micro groundwater depletion and optimizing water extraction. The framework integrates satellite remote sensing, atmospheric predictors, subsurface physical profiles, and in-situ monitoring data.

Key Data Sources & Assets

- **Macro-Scale Gravimetry**: Uses NASA GRACE/GRACE-FO JPL RL06.3Mv04 Mascon grids to measure global Terrestrial Water Storage Anomalies ($TWSA$). Complementary datasets from CSR and GFZ provide ensemble validation to model structural gravimetric uncertainty.
- **Hydroclimatic & Atmospheric Predictors**:
  - **Precipitation (Pr)**: CHIRPS v2.0/v3.0 provides 0.05° (~5 km) daily/monthly rainfall inputs.
  - **Evapotranspiration (ET)**: MODIS (MOD16A2) provides 500-meter 8-day composite actual ET. WaPOR provides localized 30–250 m resolution across Africa and the Near East.
- **Land Surface Models (LSM)**: GLDAS v2.1 (NOAH/CLSM/VIC) supplies Soil Moisture ($SMSA$), Snow Water Equivalent ($SWEA$), and Canopy Water ($CWSA$) anomalies.
- **Static Subsurface & Topography**:
  - **Soil Properties**: ISRIC SoilGrids 250m v2.0 provides soil texture (clay, sand, silt) across six depth layers to derive specific yield (S_y) maps via pedotransfer functions.
  - **Depth to Bedrock (DTB)**: Shangguan et al./EarthEnv layers establish vadose zone thickness to parameterize time-lag kernels (κ).
  - **Geomorphology**: EarthEnv Topography suite supplies 90m/250m elevation, slope, aspect, and terrain indices.
- **In-Situ Observational Networks & Pumping Data**:
  - **Ground-Truth Target Data**: Integrates USGS NWIS (US), EPA HydroNet (Europe), and CGWB (India) piezometric well networks for model training and evaluation.
  - **Anthropogenic Extraction ($E_{i(t)}$)**: Meter logs, permit registries, or satellite-derived agricultural energy balance proxies model active pumping stresses.

Data Pipeline & Preprocessing Architecture





1. **Spatial Harmonization & Mass Conservation**: Reprojects inputs to a target spatial grid (e.g., 4 km). High-resolution estimates are scaled back to native GRACE mascon boundaries to enforce mass conservation across catchments.
2. **Groundwater Signal Isolation**: Subtracts shallow storage components from total water storage anomalies to isolate deep groundwater storage anomalies ($GWSA$):

$$GWSA(x, y, t) = TWSA_{\text{scaled}}(x, y, t) - SMSA(x, y, t) - SWEA(x, y, t) - CWSA(x, y, t)$$

3. **Piezometric Head Conversion**: Converts $GWSA$ into Groundwater Level Anomalies ($GWLA$) using specific yield (S_y) maps derived from SoilGrids:

$$GWLA(x, y, t) = \frac{GWSA(x, y, t)}{S_{y(x,y)}}$$

4. **Vadose Zone Delay Transport**: Passes net surface fluxes ($P - ET$) through a gamma-convolution transfer function parameterized by Depth to Bedrock (DTB) and clay content to calculate lagged effective recharge (R_{eff}).

Model Integration Formats

- **Graph Neural Networks (GNNs)**: Formats well networks into directed graphs $G = (V, E)$, where nodes represent spatial temporal dynamics (R_{eff}, $E_{i(t)}$, G_{obs}) and edges capture transmissivity and hydraulic connectivity.
- **Fourier Neural Operators (FNOs)**: Encodes hydroclimatic variables into spatial 2D spectral representations using FFTs for rapid, resolution-independent evaluations.
- **Chance-Constrained Model Predictive Control (CCMPC)**: Converts states into discrete-time matrices for Sequential Convex Programming (SCP) solvers to compute extraction limits ($E_{i(t)}$) while preventing water table levels from falling below critical thresholds ($G_{\min}$).

User prompt: just give me the detailed summary of only data sources

Response: Comprehensive Summary of Hydroinformatic Data Sources

1. Macro-Scale Satellite Gravimetry

- **NASA GRACE & GRACE-FO Mascon Grids**
  - **Provider / Authority**: NASA Jet Propulsion Laboratory (JPL) and Physical Oceanography Distributed Active Archive Center (PO.DAAC).
  - **Spatial Resolution**: $0.25^\circ \times 0.25^\circ$ global grid (derived from 4,551 native equal-area 3° spherical cap mascons, $\sim 300\ \mathrm{km}$ footprint).
  - **Temporal Span & Cadence**: Monthly time-series from March 31, 2002, to present.
  - **Target Variables**: Equivalent Liquid Water Thickness Anomalies ($TWSA$, in cm) with cell-specific 1σ observational uncertainties.
  - **Access Handle**: Google Earth Engine snippet `ee.ImageCollection("NASA/GRACE/MASS_GRIDS_V04/MASCON")` or NetCDF files via DOI `10.5067/TEMSC-3MJ634`.
  - **Role in Pipeline**: Provides the macro-scale total column water mass change constraint ($TWSA_{\text{native}}$) to ensure regional mass conservation during downscaling.
- **Multi-Center Ensemble Streams (CSR & GFZ)**
  - **Provider / Authority**: Center for Space Research (CSR) at UT Austin and GFZ German Research Centre for Geosciences.
  - **Target Variables**: Level-2 and Level-3 spherical harmonic and mascon variations.
  - **Role in Pipeline**: Functions as an ensemble baseline to quantify and reduce structural gravimetric uncertainties before deep learning downscaling.

2. Hydroclimatic and Atmospheric Predictor Streams

- **CHIRPS Precipitation (Pr)**
  - **Provider / Authority**: Climate Hazards Center (CHC), UC Santa Barbara.
  - **Spatial Resolution**: $0.05^\circ \times 0.05^\circ$ ($\sim 5\ \mathrm{km}$) spanning $60^\circ\mathrm{N}$ to $60^\circ\mathrm{S}$.
  - **Temporal Span & Cadence**: Continuous daily, pentadal, dekadal, and monthly coverage from 1981 to near-present.
  - **Target Variables**: Gridded daily precipitation depth (Pr, in mm/day).
  - **Access Handle**: Google Earth Engine snippet `ee.ImageCollection("UCSB-CHG/CHIRPS/DAILY")` or direct HTTP GeoTIFF/NetCDF downloads.
  - **Role in Pipeline**: Supplies time-varying infiltration mass input $P(t)$ for the lagged gamma-convolution transfer function and spatial downscaling regressors.
- **Evapotranspiration (ET) Products**
  - **Primary Dataset (MODIS MOD16A2 v6.1)**:
    - **Provider / Authority**: NASA Land Processes Distributed Active Archive Center (LP DAAC).

        - **Spatial/Temporal Resolution**: 500-meter pixel resolution aggregated in 8-day composite intervals globally.
        - **Target Variables**: Actual Evapotranspiration (ET), Potential Evapotranspiration (PET), and Latent Heat Flux.
        - **Access Handle**: Google Earth Engine asset `ee.ImageCollection("MODIS/061/MOD16A2")`.
    - **Regional Complement (FAO WaPOR v2.0/v3.0)**:
        - **Spatial Resolution**: High-resolution $30\ \text{m}$ to $250\ \text{m}$ across Africa and the Near East.
    - **Role in Pipeline**: Provides atmospheric loss variable $ET(t)$ needed to compute net effective surface infiltration $P(t) - ET(t)$.

3. Land Surface Model (LSM) Components

- **GLDAS v2.1 (NOAH, CLSM, VIC)**
    - **Provider / Authority**: NASA Goddard Earth Sciences Data and Information Services Center (GES DISC).
    - **Spatial & Temporal Resolution**: $0.25^\circ \times 0.25^\circ$ grid evaluated at 3-hourly and monthly timesteps.
    - **Target Variables**:
        - Soil Moisture Storage Anomaly ($SMSA$) across four root-zone depth layers (0–10 cm, 10–40 cm, 40–100 cm, and 100–200 cm).
        - Snow Water Equivalent Anomaly ($SWEA$).
        - Plant Canopy Water Storage Anomaly ($CWSA$).
        - Surface Runoff Export (Q_s).
    - **Access Handle**: Google Earth Engine asset `ee.ImageCollection("NASA/GLDAS/V021/NOAH/G300M")`.
    - **Role in Pipeline**: Enables shallow mass isolation by subtracting non-groundwater column storage components ($SMSA$, $SWEA$, $CWSA$) from total storage anomalies ($TWSA$).

4. Static Geophysical, Hydrogeological, and Subsurface Boundaries

- **ISRIC SoilGrids 250m v2.0**
    - **Provider / Authority**: ISRIC – World Soil Information (derived from WoSIS profile snapshots).
    - **Spatial & Depth Profiling**: 250-meter global grid across 6 depth intervals (0–5 cm, 5–15 cm, 15–30 cm, 30–60 cm, 60–100 cm, 100–200 cm).
    - **Target Variables**: Percentage of clay, sand, and silt; bulk density; organic carbon; coarse fragments.
    - **Access Handle**: Web Coverage Services (WCS) via `[https://maps.isric.org](https://maps.isric.org)` and Virtual Raster Tile (VRT) mosaics.

- **Role in Pipeline**: Supplies clay and sand fraction arrays to calculate unconfined specific yield (S_y) maps via pedotransfer functions, facilitating the conversion of mass storage anomalies ($GWSA$) to piezometric head changes ($GWLA$).

- **Depth to Bedrock (DTB) & Regolith Thickness**
  - **Provider / Authority**: Shangguan et al., integrated with EarthEnv geomorphological layers.
  - **Spatial Domain**: 250-meter global grid compiled from ~130,000 soil profiles and ~1.6 million borehole records.
  - **Target Variables**: Absolute Depth to Bedrock (DTB, in cm) and R-horizon occurrence probability.
  - **Access Handle**: Distributed via `[http://www.earthenv.org/](http://www.earthenv.org/)`.
  - **Role in Pipeline**: Parameterizes shape (α_{lag}) and scale (β_{lag}) factors in the gamma-distribution lag kernel to accurately simulate physical vadose zone percolation delays.

- **Digital Elevation & Geomorphology (EarthEnv / SRTM v4.1 / GMTED2010)**
  - **Provider / Authority**: Yale University / NASA JPL.
  - **Spatial Resolution**: 90-meter and 250-meter global rasters.
  - **Target Variables**: Elevation, Slope Angle, Aspect, Terrain Roughness Index (TRI), Topographic Position Index (TPI), Curvature, and Landform Classes.
  - **Access Handle**: Downloadable at `[http://www.earthenv.org/topography](http://www.earthenv.org/topography)`.
  - **Role in Pipeline**: Provides static terrain covariates $S(x, y)$ for downscaling and establishes boundary conditions for Physics-Informed Neural Networks (PINNs).

5. In-Situ Piezometric Wells & Anthropogenic Extraction Registries

- **Ground-Truth Well Networks**
  - **USGS NWIS (United States)**: Accessible programmatically using Python `dataretrieval` (`nwis.get_gwlevels`); provides historical manual/automated piezometric head measurements.
  - **EPA HydroNet & North German Basin Database (Europe)**: Over 3,000 reference observation wells accessed via national environmental portals.
  - **CGWB WRAMIS Portals (India)**: Pre- and post-monsoon piezometric monitoring data across dense agricultural networks.
  - **Role in Pipeline**: Establishes ground-truth target vectors (G_{obs}) to train, validate, and evaluate performance metrics (RMSE/MAE) of machine learning surrogates.

- **Anthropogenic Volumetric Extraction Data ($E_{i(t)}$)**
  - **Sources**: Agricultural flow-meter databases, municipal utility registries, and water allocation permits. Unmetered basins use thermal Landsat 8/9 surface energy balance models coupled with crop water demand functions as indirect proxies.
  - **Role in Pipeline**: Serves as explicit control variable inputs ($E_{i(t)}$) optimized within the Chance-Constrained Model Predictive Control (CCMPC) pipeline.

User prompt: summarise this again

Response: Data Sources Summary

1. Gravimetric Remote Sensing

- **NASA GRACE & GRACE-FO Mascon Grids**: Global $0.25^\circ \times 0.25^\circ$ monthly grids (derived from 4,551 equal-area 3° mascons) tracking Terrestrial Water Storage Anomalies ($TWSA$, in cm). Provides total water mass constraints. Accessed via Google Earth Engine (`ee.ImageCollection("NASA/GRACE/MASS_GRIDS_V04/MASCON")`) or NASA Earthdata (DOI `10.5067/TEMSC-3MJ634`).
- **CSR & GFZ Ensemble Solutions**: Spherical harmonic and mascon variations from UT Austin (CSR) and GFZ Potsdam. Used to estimate structural uncertainties in satellite gravity estimates.

2. Hydroclimatic Surface Inputs

- **CHIRPS Precipitation (v2.0/v3.0)**: Quasi-global (60°N–60°S) daily and monthly rainfall estimates at 0.05° (~5 km) spatial resolution. Functions as the primary mass input $P(t)$ for vadose zone infiltration models. Accessed via `ee.ImageCollection("UCSB-CHG/CHIRPS/DAILY")`.
- **MODIS & FAO WaPOR Evapotranspiration**: MODIS (MOD16A2 v6.1) supplies global 500-meter, 8-day composite actual ET data via `ee.ImageCollection("MODIS/061/MOD16A2")`. FAO WaPOR provides localized 30–250 meter ET grids for Africa and the Near East. Defines atmospheric water losses to calculate net infiltration ($P - ET$).

3. Land Surface Models

- **GLDAS v2.1 (NOAH/CLSM/VIC)**: 0.25° 3-hourly and monthly grids from NASA GES DISC. Supplies Soil Moisture ($SMSA$), Snow Water Equivalent ($SWEA$), Canopy Water ($CWSA$), and Surface Runoff (Q_s). Accessed via `ee.ImageCollection("NASA/GLDAS/V021/NOAH/G300M")`. These components are subtracted from $TWSA$ to isolate deep groundwater storage ($GWSA$).

4. Subsurface & Geophysical Boundaries

- **ISRIC SoilGrids 250m v2.0**: 250-meter global grids across six depth layers (0–200 cm) providing sand, clay, silt fractions, and bulk density. Accessed via WCS (`[https://maps.isric.org](https://maps.isric.org)`). Used to compute specific yield (S_y) maps for converting storage anomalies ($GWSA$) to piezometric level changes ($GWLA$).
- **Depth to Bedrock (DTB)**: 250-meter global grid compiled from ~1.6 million borehole logs and 130,000 soil profiles. Configures shape (α_{lag}) and scale (β_{lag}) parameters in gamma-convolution kernels to model vadose zone transit delays. Available via `[http://www.earthenv.org/](http://www.earthenv.org/)`.
- **EarthEnv Topography**: Global 90m/250m DEM rasters supplying elevation, slope, aspect, and terrain indices (TRI, TPI). Used as spatial covariates and neural network boundary constraints.

5. Ground-Truth & Pumping Data

- **Observation Well Networks**: Piezometric water table time-series used for target training and verification (G_{obs}). Integrates USGS NWIS (accessed via Python `dataretrieval`), European EPA HydroNet/North German Basin datasets (>3,000 wells), and India's CGWB WRAMIS portal.
- **Anthropogenic Extraction Registries**: Volumetric pumping rates ($E_{i(t)}$) sourced from flow-meters, utility records, and allocation permits. Unmetered areas use thermal Landsat 8/9 surface energy balance models. Serves as the control input for extraction optimization.

User prompt: give me the datasouces with direct link to download

Response: Here are the data sources from the catalog along with their direct access links, programmatic endpoints, or repository portals:

1. Macro-Scale Satellite Gravimetry

- **NASA GRACE & GRACE-FO Mascon Grids**
  - **Direct HTTP / PO.DAAC Portal**: NASA Earthdata PO.DAAC Repository (DOI: 10.5067/TEMSC-3MJ634)[cite: 1]
  - **Google Earth Engine Endpoint**: `ee.ImageCollection("NASA/GRACE/MASS_GRIDS_V04/MASCON")`[cite: 1]

2. Hydroclimatic & Atmospheric Predictors

- **CHIRPS Precipitation (Pr)**
  - **Direct HTTP Download Portal**: UCSB Climate Hazards Center Data Repository[cite: 1]
  - **Google Earth Engine Endpoint**: `ee.ImageCollection("UCSB-CHG/CHIRPS/DAILY")`[cite: 1]
- **MODIS Global Evapotranspiration (MOD16A2 Version 6.1)**
  - **Direct Access Portal**: NASA Earthdata Search Portal[cite: 1]
  - **Google Earth Engine Endpoint**: `ee.ImageCollection("MODIS/061/MOD16A2")`[cite: 1]

3. Land Surface Model (LSM) Components

- **GLDAS v2.1 (NOAH LSM)**
  - **Direct Access Portal**: NASA GES DISC Portal[cite: 1]
  - **Google Earth Engine Endpoint**: `ee.ImageCollection("NASA/GLDAS/V021/NOAH/G300M")`[cite: 1]

4. Static Geophysical & Hydrogeological Data

- **ISRIC SoilGrids 250m v2.0**
  - **Web Coverage Service (WCS)**: ISRIC Maps WCS Portal[cite: 1]
  - **Virtual Raster Tile (VRT) Mosaics**: ISRIC File Storage Repository[cite: 1]
- **Global Depth to Bedrock (DTB)**

- **Direct Download Portal**: EarthEnv Soil & Depth to Bedrock Layer Downloads[cite: 1]

- **EarthEnv Topography Suite (Elevation, Slope, Aspect, TRI, TPI)**
  - **Direct Download Portal**: EarthEnv Topography Download Page[cite: 1]

5. In-Situ Monitoring Networks

- **USGS National Water Information System (NWIS)**
  - **Programmatic Retrieval (Python)**: Official `dataretrieval` package[cite: 1]

    ```
    from dataretrieval import nwis
    # Use nwis.get_gwlevels() or nwis.get_dv() functions
    ```

  - **Web Portal**: USGS NWIS Web Interface[cite: 1]
